

NYSE Pillar Market-Data Engine

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1 Introduction

Potential project structure:

```
project/  
  README.md  
  src/  
  tests/  
    unit/  
    integration/  
    parsing/  
  examples/  
    replay/  
    stock-summary/  
    bbo-validation/  
  data/  
    input.csv
```

2 Engine

At its core, the engine should be a decoder and a source of heuristics. This abstracts away much of the complexity involved in working with the data, allowing the engine to relay decoded messages to a consumer as quickly as possible.

Characteristics that the engine could or should track include `msg/sec`, gap detection, and the specific messages decoded.

Features that the engine could provide may include viewing specific messages and replaying a selected interval of messages.

3 Validation and Testing Methods

Could use TAQ NYSE BBO to validate order book data that will eventually be derived from the engine. Also should use STOCKSUM data to verify.